

Soham Amol Dube

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Education

B.Tech in Computer Science Engineering (AI and ML), VIT Bhopal University; CGPA: 8.61 Aug 2023 – May 2027
Higher Secondary (12th), Percentage: 86.5% February 2023
Secondary (10th), Percentage: 86.2% August 2021

Technical Skills

Programming: Python, SQL, JavaScript/TypeScript
Software Testing & Quality: Unit & Integration Testing, API Testing (Postman), CI/CD Test Automation, Regression Testing, Debugging & Root-Cause Analysis
AI/ML & Big Data: PyTorch, LangChain, Transformers, LSTMs, NLP, Apache Spark, Pandas, NumPy, scikit-learn
Full-Stack & Cloud: React, Node.js, Express, MongoDB, FastAPI, HTML/CSS, AWS (EC2, S3), REST API Design
Infrastructure & DevOps: Docker, Git, GitHub Actions, Linux, SQLite

Internship

MERN Stack Developer Intern – IIT Ropar (Prof. Sudarshan Iyengar) Dec 2025 – Jan 2026

- Architected a modular full-stack MERN solution from initial case study requirements, delivering a highly scalable digitization workflow for academic administration used across multiple departments.
- Designed and validated RESTful APIs, writing test cases against expected request/response contracts and optimizing MongoDB queries with indexing and aggregation pipelines, reducing server response time by 25% under load.
- Implemented secure JWT authentication and role-based access control; structured the React/Node.js codebase for CI/CD integration and verified builds via automated pipeline checks before AWS production deployment.

Projects

GenAI Client Integrator (Full-Stack RAG)

- Architected a full-stack Retrieval-Augmented Generation application using FastAPI and React to automate complex document analysis, summarization, and structured data extraction using LLMs.
- Integrated ChromaDB for efficient vector storage, embedding generation, and semantic retrieval, reducing manual document review time by over 60% for end users.
- Dockerized the entire infrastructure using `docker-compose` with isolated services for the API, vector store, and frontend, enabling reproducible service-level testing and seamless deployment across environments.

Live Data Pipeline & Spark Dashboard

- Built a resilient data ingestion engine using Python and PySpark to fetch, clean, transform, and store high-velocity REST API data streams in a SQL database at scale.
- Engineered automated ETL workflows with fault tolerance, retry logic, and structured error handling, validated through simulated network-interruption and upstream-failure test scenarios to ensure zero data loss.
- Developed an interactive Plotly Dash visualization layer with filterable, real-time charts to serve operational insights derived from the ingested big data pipeline.

AI-Based Stock Prediction (Spark + LSTM)

- Engineered a distributed data pipeline using Apache Spark to process massive historical stock datasets and scale complex feature engineering across multiple tickers.
- Built and validated a PyTorch LSTM model using 10-time-step sequences and hyperparameter tuning, benchmarking predictive stability and directional accuracy across varying market conditions.

Context Handoff Protocol — AI-Powered Team Handoff Briefs | *Python, FastAPI, HDBSCAN, LLMs*

- Built a full-stack AI pipeline that ingests cross-platform team signals from Slack, GitHub, Notion, and email to generate structured handoff briefs for engineering and product teams.
- Engineered a modular four-stage pipeline for ingestion, HDBSCAN-based semantic clustering, staleness scoring, and LLM synthesis, with provider-agnostic embedding and LLM interfaces configurable without code changes.
- Developed an asynchronous FastAPI backend with SQLAlchemy/SQLite persistence, validating API endpoints and pipeline stages against both real and synthetic data sources via pluggable test connectors; deployed on HuggingFace Spaces.

Certifications

- IBM Generative AI Intern** May 2025 – Jun 2025
- Applied Machine Learning in Python** Dec 2024

Extracurricular Activities

Tech Team Lead – Health-O-Tech Jan 2025 – Present

- Led a 4-member cross-functional team to orchestrate the 2-day ‘Run Riot’ initiative for 300+ students, integrating a Luma-based registration system to streamline onboarding and check-in.
- Contributed to the architecture and full-stack development of the official club website to centralize operations, manage events, and drive user engagement across the student community.